

	$= 9.58 \times 10^6 \text{ J}$	
4(a)(i)	$\Delta\phi = \frac{\Delta x}{\lambda} \times 2\pi$ $\frac{3}{2}\pi = \frac{0.490}{\lambda} \times 2\pi$ $\lambda = 0.653 \text{ m}$ $v = f\lambda$ $= (520)(0.653)$ $= 340 \text{ m s}^{-1}$	
4(a)(ii)	$I = \frac{P}{4\pi r^2}$ Since P is constant, $I \propto \frac{1}{r^2}$ $\frac{I_Y}{I_X} = \frac{r_X^2}{r_Y^2}$ $I_Y = \frac{(5.24)^2}{(5.73)^2} (2.6 \times 10^{-3})$ $= 2.17 \times 10^{-3} \text{ W m}^{-2}$	
4(b)(i)	$Amp = Amp_0 \cos \theta$ $\frac{A}{2} = A \cos \theta$ $\cos \theta = \frac{1}{2}$ $\theta = 60^\circ$	
4(b)(ii)	$I = kAmp^2$ fractional change $= \frac{I_0 - I}{I_0}$ $= \frac{kA^2 - k\left(\frac{A}{2}\right)^2}{kA^2}$ $= 1 - \frac{\left(\frac{A}{2}\right)^2}{A^2} = 0.75$	

5(a)	n.d. across $P = 1.3 \text{ V}$	
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